

GRANT PETRI

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EDUCATION

The University of Texas at San Antonio *San Antonio, TX | Bachelor of Science in Computer Science* May 2026 | GPA: 3.6 • President's List

Game Development Coursework: Game Programming, Engine Programming, UI/UX Design, Data Structures & Algorithms, Parallel Programming

TECHNICAL SKILLS

Engines & Tools: Unity, Roblox Studio, Blender, Godot, Git, Docker

Languages: C#, Lua, Python, C, Go, Java

Core Strengths: Gameplay Systems Architecture, Physics Simulation, Procedural Generation, AI Navigation & Pathfinding, Networked Multiplayer Systems, 3D Modeling & Animation

GAME DEVELOPMENT PROJECTS

Multiplayer FPS Framework *Roblox Studio | Lua*

- Built a full multiplayer FPS framework from the ground up, including server-authoritative validation to prevent client-side exploits and keep gameplay fair across all connected clients.
- Built a modular weapon/tool system supporting multiple distinct tool types, each driven by dedicated state machines to manage, equip, fire, reload, and inspect behavior cleanly.
- Designed first-person view models and custom animations, integrating simulated spring physics to drive realistic weapon sway and recoil.
- Replicated gameplay state across clients in real time, balancing responsiveness with server authority.

Sandbox Placement & Inventory Systems *Roblox Studio | Lua*

- Built a free-form sandbox object placement system supporting flexible, player-driven world building.
- Designed and implemented inventory systems for managing, storing, and equipping in-game items.
- Developed AI navigation systems enabling NPCs to path intelligently through dynamic environments.

Procedural Generation & Physics Simulation *Unity | C#*

- Designed and implemented multiple procedural generation systems across both 2D and 3D projects, using parameterized rulesets to dynamically build maps and visual compositions.
- Built a custom 2D physics simulation from scratch, including an adjustable spring system for realistic deformable-body behavior.
- Developed intelligent AI navigation and pathfinding for context-aware NPC movement.
- Built a first-person viewmodel system for weapon/tool presentation.

3D Art & Animation *Blender*

- Modeled and rigged original characters, objects, and environment maps; created custom animations to bring rigged characters to life.
- Intermediate proficiency across modeling, rigging, and animation workflows, applied directly to game project assets.

EXPERIENCE

Ronin POS — Quality Assurance *San Antonio, TX | July 2023 – December 2024*

- Executed manual and regression testing on point-of-sale software, identifying and documenting defects across builds to support reliable releases.

CERTIFICATIONS

Microsoft Intune & Azure • Microsoft Excel • Entrepreneurship and Small Business (ESB) • IRS VITA Certified Volunteer Tax Preparer (2019–2020)